

Contribution Journal — Module 04

Module 04 Contribution Journal — Exploratory Data Analysis

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Course: ITAI-1371

For this lab, my individual work focused on completing and documenting the Exploratory Data Analysis notebook. I reviewed the notebook instructions and followed the workflow from loading the Titanic dataset to examining summary statistics and visualizations.

I completed the two student experimentation sections. In the first experiment, I added a count plot to compare survival outcomes by port of embarkation and calculated survival proportions for each port. In the second experiment, I added a box plot to compare ticket fares for passengers who survived and passengers who did not. I included titles and axis labels so the charts would be easier to understand, along with short interpretation statements explaining what to look for.

I also completed the three knowledge-check answers in my own words. My answers explain the purpose of EDA, summarize the patterns described by the notebook's class, gender, and age plots, and explain why visualizations are useful alongside summary statistics. I tried to avoid treating a pattern as proof of cause and effect.

Another part of my contribution was preparing the reflection and organizing the notebook content for submission. The reflection describes what I learned about using charts and statistics together, selecting a plot for different data types, and questioning assumptions about relationships in data.

Remaining verification step: The notebook retrieves the Titanic dataset from an online source. In this working environment, the data URL was not reachable, so I could not verify the chart outputs or claim that every cell executed successfully. I need to open the completed notebook in Google Colab, run all cells, review the results, and export the final executed notebook to PDF. I will make any corrections if the real outputs differ from the expected interpretations.

This journal describes the notebook work prepared for this submission. I will edit it if any detail does not match the work I personally completed.